JEE Mathematics: Quick Review with Grading and Practice Map - Vector 3D (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (250 questions; a question that appears in two of these is counted once): High = 23 or more questions, Medium = 1 to 22 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Introduction

Question 01

Define vector, initial point, terminal point, head and tail of a vector

Model Answer

A vector is a directed line segment (magnitude + direction, obeys vector addition). In $\overrightarrow{AB}$, A = initial point (tail), B = terminal point (head).
$\overrightarrow{AB} \neq \overrightarrow{BA}$: the end points cannot be interchanged; $\overrightarrow{BA} = -\overrightarrow{AB}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 02

What is the direction of a vector?

Model Answer

The direction from the tail to the head along its line of support; it is given by the unit vector $\hat{a} = \dfrac{\vec{a}}{|\vec{a}|}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 03

What is the magnitude of a vector?

Model Answer

The length of the directed segment: $|\overrightarrow{AB}| = AB \geq 0$, and $|\overrightarrow{AB}| = |\overrightarrow{BA}|$.

Grade - Low

Practice map, main concept: none

Also used in: DPP: 3(6)

Question 04

What is line of support of a vector?

Model Answer

The line of unlimited length of which the directed segment is a part (also called the support).

Grade - Low

Practice map, main concept: none

Also used in: none

Important Definitions

Question 05

Define null vector and how is it represented?

Model Answer

A vector of magnitude zero, written $\vec{0}$; initial and terminal points coincide, its direction is arbitrary.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 06

Define unit vector and how is it represented?

Model Answer

A vector of magnitude 1. Unit vector along $\vec{a}$: $\hat{a} = \dfrac{\vec{a}}{|\vec{a}|}$; $\hat{i}, \hat{j}, \hat{k}$ are along the axes.

Ex: $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k} \Rightarrow \hat{a} = \frac{1}{3}(2\hat{i} - \hat{j} + 2\hat{k})$.

Grade - Medium

Practice map, main concept: Question Bank: 37

Also used in: Sheet: J-I 8; J-II 5, 53, 56; J-M 29 | Question Bank: 1, 32-33

Question 07

How to find magnitude of a vector expressed in $\hat{i}, \hat{j}, \hat{k}$ form?

Model Answer

$$|x\hat{i} + y\hat{j} + z\hat{k}| = \sqrt{x^2 + y^2 + z^2}$$

Ex: $|3\hat{i} - 4\hat{j} + 12\hat{k}| = 13$. For points: $AB = |\vec{b} - \vec{a}|$.

Grade - Medium

Practice map, main concept: Sheet: J-I 2 | DPP: 1(3-5) | Question Bank: 29

Also used in: Sheet: J-I 1; J-II 58; J-M 12, 45-46, 85, 90, 96; J-A 3, 9, 49; C-I 15; C-II 2, 5, 28, 33 | DPP: 1(1, 6, 10); 6(16); 7(7, 10); 9(8) | Question Bank: 28, 41, 44-46, 53, 56, 62, 69

Question 08

What is the role of unit vector?

Model Answer

It gives only the direction. A vector of magnitude k along $\vec{a}$ is $k\hat{a}$ (opposite: $-k\hat{a}$).

Ex: magnitude 6 along $2\hat{i} - \hat{j} + 2\hat{k}$: $4\hat{i} - 2\hat{j} + 4\hat{k}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 56

Also used in: Sheet: J-I 26; J-II 15; J-M 13; C-II 7 | DPP: 9(9); 10(12)

Question 09

When are two vectors equal?

Model Answer

When they have the same magnitude and the same direction (their position in space does not matter).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 10
How to equate two vectors expressed in $\hat{i}, \hat{j}, \hat{k}$ form?

Model Answer
Compare the coefficients: $a_1 = b_1,\ a_2 = b_2,\ a_3 = b_3$.
Ex: $(x+y)\hat{i} + (x-y)\hat{j} + z\hat{k} = 5\hat{i} + \hat{j} - 2\hat{k} \Rightarrow x = 3,\ y = 2,\ z = -2$.

Grade - Medium
Practice map, main concept: Sheet: C-II 18, 22
Also used in: Sheet: J-II 10 | Question Bank: 24

Question 11
How is the angle between two vectors defined and how is angle denoted?

Model Answer
The smaller angle θ formed when the tails (or the heads) of the two vectors are brought together; written $\theta = (\widehat{\vec{a}, \vec{b}})$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 12
What is the range of angle between two vectors?

Model Answer

$$0 \leq \theta \leq \pi$$

$\theta = 0$: like parallel; $\theta = \pi$: unlike parallel; $\theta = \frac{\pi}{2}$: perpendicular.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 13
How to calculate angle if head and tail are placed together?

Model Answer
Shift one vector so that both start from the same point; if the figure angle is α between head-to-tail vectors, the required angle is $\pi - \alpha$.
Ex: equilateral $\triangle ABC$: angle between $\overrightarrow{AB}$ and $\overrightarrow{BC}$ is $120°$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 14
What do you mean by collinear vectors?

Model Answer
Vectors whose lines of support are parallel or the same (also called parallel vectors), whatever their directions.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 15
How to check whether two collinear vectors are like or unlike vectors?

Model Answer
Write $\vec{a} = \lambda\vec{b}$: $\lambda > 0$ like (same direction), $\lambda < 0$ unlike (opposite).
Ex: $2\hat{i} - 4\hat{j} + 6\hat{k} = -2(-\hat{i} + 2\hat{j} - 3\hat{k})$: unlike.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 16
What is the condition of two vectors, expressed in $\hat{i}, \hat{j}, \hat{k}$ form, to be collinear?

Model Answer

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} \ (= \lambda)$$

Points A, B, C are collinear $\Leftrightarrow \overrightarrow{AB} = \lambda\overrightarrow{AC}$.
Ex: $(x+2)\hat{i} - (x-y)\hat{j} + \hat{k} \parallel (x-1)\hat{i} + (2x+y)\hat{j} + 2\hat{k} \Rightarrow x = -5,\ y = -20$.

Grade - Medium
Practice map, main concept: Sheet: J-II 2; J-M 61; J-A 33; C-II 25 | DPP: 1(7); 10(5)
Also used in: Sheet: J-II 8, 62 | DPP: 2(3) | Question Bank: 11, 67

Question 17
How to assume a vector parallel to a given vector $\vec{p}$?

Model Answer
Take it as $\lambda\vec{p}$, $\lambda \in \mathbb{R}$; then use the extra condition (magnitude, dot product...) to find λ.

Grade - Medium
Practice map, main concept: DPP: 3(6)
Also used in: Sheet: J-I 4 | DPP: 4(3) | Question Bank: 25, 55

Question 18
What do you mean by coplanar vectors?

Model Answer
Vectors whose directed segments lie in (or are parallel to) one plane.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 19
How to check coplanarity of two vectors?

Model Answer
No check is needed: any two vectors are always coplanar (bring their tails together). Three vectors need $[\vec{a}\ \vec{b}\ \vec{c}] = 0$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 20
What to keep in mind while shifting vectors in Mathematics?

Model Answer
Vectors are free: they may be shifted anywhere keeping magnitude and direction unchanged. Localised vectors (e.g. force at a point) are not shifted.

Grade - Low
Practice map, main concept: none
Also used in: none

Addition of vectors

Question 21
How to write relation between vectors forming a closed figure?

Model Answer
Taken head to tail around a closed figure, the sum is $\vec{0}$: $\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CA}=\vec{0}$ (triangle law $\overrightarrow{AB}+\overrightarrow{BC}=\overrightarrow{AC}$).
Ex: if C is the mid-point of AB, then $\overrightarrow{PA}+\overrightarrow{PB}=2\overrightarrow{PC}$ for any P.

Grade - Medium
Practice map, main concept: Sheet: J-I 3; J-M 91; C-I 16 | DPP: 2(3)
Also used in: Sheet: J-I 10, 15; J-A 18, 36 | DPP: 1(6, 9); 8(2) | Question Bank: 38

Question 22
How to write diagonal vectors of a parallelogram whose adjacent side vectors are $\vec{a}$ and $\vec{b}$?

Model Answer

$$\vec{d_1}=\vec{a}+\vec{b}, \quad \vec{d_2}=\vec{b}-\vec{a}$$

Also $|\vec{a}+\vec{b}|^2+|\vec{a}-\vec{b}|^2=2(|\vec{a}|^2+|\vec{b}|^2)$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-I 2, 6; J-A 6; C-II 11, 21 | DPP: 3(4); 5(3)

Question 23
How to add two vectors expressed in $\hat{i},\hat{j},\hat{k}$ form?

Model Answer
Add the corresponding components: $\vec{a}+\vec{b}=(a_1+b_1)\hat{i}+(a_2+b_2)\hat{j}+(a_3+b_3)\hat{k}$.
Ex: $(\hat{i}+2\hat{j}-\hat{k})+(2\hat{i}-\hat{j}+3\hat{k})=3\hat{i}+\hat{j}+2\hat{k}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 24
What do you mean by additive inverse?

Model Answer
$-\vec{a}$: same magnitude, opposite direction, with $\vec{a}+(-\vec{a})=\vec{0}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 25
Write expression of triangle inequality.

Model Answer

$$\big||\vec{a}| - |\vec{b}|\big| \leq |\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$$

Ex: $|\vec{a}| = 3,\ |\vec{b}| = 4 \Rightarrow 1 \leq |\vec{a} + \vec{b}| \leq 7$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 26
What is the condition on two vectors in case of equality in triangle inequality?

Model Answer
$|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}|$ iff $\vec{a}, \vec{b}$ are like parallel; $|\vec{a} + \vec{b}| = \big||\vec{a}| - |\vec{b}|\big|$ iff unlike parallel (or one is $\vec{0}$).

Grade - Low
Practice map, main concept: none
Also used in: none

Multiplication of a vector by scalar

Question 27
What is the effect on magnitude of a vector if a vector gets multiplied by λ?

Model Answer

$$|\lambda\vec{a}| = |\lambda|\,|\vec{a}|$$

Grade - Low
Practice map, main concept: none
Also used in: none

Question 28
What is the effect on direction of a vector if a vector gets multiplied by λ?

Model Answer
$\lambda > 0$: same direction; $\lambda < 0$: opposite direction; $\lambda = 0$: null vector.
Ex: $-2\vec{a}$ has twice the length of $\vec{a}$ and the opposite direction.

Grade - Low
Practice map, main concept: none
Also used in: none

Coordinate System in 3D

Question 29

How to define x, y and z coordinates of a point?

Model Answer

x, y, z = signed distances of P from the yz-, zx- and xy-planes; position vector $\overrightarrow{OP} = x\hat{i} + y\hat{j} + z\hat{k}$. The three coordinate planes divide space into 8 octants.

Ex: origin shifted to $(1, 2, -3)$ (axes parallel): $(0, 4, 5)$ becomes $(-1, 2, 8)$.

Grade - Medium

Practice map, main concept: Sheet: J-II 3-4; C-I 14 | DPP: 1(2, 6); 2(7); 6(7) | Question Bank: 21, 44-46, 56

Also used in: Sheet: J-I 1, 3; J-II 9, 49, 57, 69; J-M 61, 89; J-A 29, 48, 50, 55; C-I 11; C-II 31-32 | DPP: 1(1, 10); 2(1, 8); 4(9, 13); 8(14); 9(2) | Question Bank: 20, 51, 53-54

Question 30

What is the condition on coordinates if a point $P(a, b, c)$ lies on x-axis / y-axis / z-axis?

Model Answer

x-axis: $b = c = 0$; y-axis: $a = c = 0$; z-axis: $a = b = 0$. Planes: xy: $c = 0$, yz: $a = 0$, zx: $b = 0$.

Grade - Medium

Practice map, main concept: none

Also used in: DPP: 6(7); 7(7) | Question Bank: 29

Question 31

How to find the distance of a point from the coordinate axes and planes, and its foot of perpendicular on them?

Model Answer

For $P(a, b, c)$: distance from x-axis $\sqrt{b^2 + c^2}$ (foot $(a, 0, 0)$), from y-axis $\sqrt{a^2 + c^2}$, from z-axis $\sqrt{a^2 + b^2}$.

From xy-plane $|c|$ (foot $(a, b, 0)$), from yz-plane $|a|$, from zx-plane $|b|$.

Ex: $P(3, 4, 12)$: from x-axis $4\sqrt{10}$, from z-axis 5, from xy-plane 12.

Grade - Medium

Practice map, main concept: Question Bank: 36

Also used in: Sheet: J-II 33; J-A 31; C-I 18 | DPP: 6(5)

Linear Dependence and Linear Combination in a Plane

Question 32

What are linearly dependent / independent vectors, and when can coefficients be compared?

Model Answer

$\vec{x}_1, \ldots, \vec{x}_n$ are independent if $k_1\vec{x}_1 + \cdots + k_n\vec{x}_n = \vec{0} \Rightarrow$ all $k_i = 0$; otherwise dependent (one is a combination of the others).

Coefficients may be compared only along independent vectors. Two collinear or three coplanar vectors are dependent; four vectors in space are always dependent.

Points with p.v. $\vec{a}, \vec{b}, \vec{c}$ are collinear iff $x\vec{a} + y\vec{b} + z\vec{c} = \vec{0}$ with $x + y + z = 0$ (not all zero).

Grade - Medium

Practice map, main concept: Sheet: J-I 4; J-II 1, 63; C-II 20 | DPP: 4(3, 8)

Also used in: Sheet: J-I 32-33; J-II 2-4; J-M 100; J-A 17; C-I 16; C-II 13, 18 | DPP: 2(8)

Question 33
What is the linear combination theorem in a plane?

Model Answer
If $\vec{a}, \vec{b}$ are non-collinear, every vector $\vec{r}$ coplanar with them is uniquely $\vec{r} = x\vec{a} + y\vec{b}$; a vector in the plane of $\vec{a}, \vec{b}$ is taken as $x\vec{a} + y\vec{b}$.
Ex: $\hat{i} + 3\hat{j} + 2\hat{k} = (\hat{i} + \hat{j}) + 2(\hat{j} + \hat{k})$.

Grade - Medium
Practice map, main concept: Sheet: J-M 3, 33, 54; C-I 13; C-II 7, 15, 30, 41 | DPP: 4(2) | Question Bank: 19
Also used in: Sheet: J-I 22; J-M 51

Vector Equation of a Line

Question 34
What is the vector equation of a line?

Model Answer

$$\vec{r} = \vec{a} + t\vec{b} \qquad \vec{r} = \vec{a} + t(\vec{b} - \vec{a})$$

Through point $\vec{a}$ parallel to $\vec{b}$; second form: through points $\vec{a}, \vec{b}$ (t = parameter).
Ex: through $(1, 2, 3)$ and $(2, 0, 1)$: $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + t(\hat{i} - 2\hat{j} - 2\hat{k})$.

Grade - Medium
Practice map, main concept: Sheet: J-II 49
Also used in: Sheet: J-II 48; J-A 32-33; C-I 20 | DPP: 2(6); 7(11) | Question Bank: 3

Question 35
How to decide whether two lines are parallel, intersecting or skew, and find their point of intersection?

Model Answer
Directions parallel: parallel (coincident if a common point exists). Otherwise equate general points, solve two component equations for t, s and test the third: satisfied = intersecting, not = skew.
Ex: $(1, 2, 3) + t(1, 1, 1)$ and $(2, 0, 1) + s(0, 2, 2)$: $t = 1,\ s = \frac{3}{2}$, meet at $(2, 3, 4)$.

Grade - Medium
Practice map, main concept: Sheet: J-II 24, 67; J-M 23, 63; J-A 9; C-II 45 | DPP: 1(8); 2(6, 8); 4(4) | Question Bank: 28, 53, 57
Also used in: Sheet: J-I 12, 35; J-II 30; J-A 11, 39, 45, 53; C-I 22 | DPP: 6(11) | Question Bank: 14, 30-31, 61

Question 36
What is the vector along the internal / external angle bisector of two vectors?

Model Answer

$$\text{internal: } \lambda(\hat{a} + \hat{b}), \qquad \text{external: } \lambda(\hat{a} - \hat{b})$$

Ex: $\vec{a} = 3\hat{i} + 4\hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$: $\hat{a} + \hat{b} = \frac{1}{15}(14\hat{i} + 10\hat{j} + 22\hat{k})$, i.e. along $7\hat{i} + 5\hat{j} + 11\hat{k}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 1, 22; J-II 11, 48; J-A 39; C-II 27 | DPP: 1(1); 9(3) | Question Bank: 23
Also used in: Sheet: J-A 38

Question 37
What is the equation of a line in cross product form?

Model Answer

$$(\vec{r}-\vec{a})\times\vec{b}=\vec{0} \iff \vec{r}\times\vec{b}=\vec{a}\times\vec{b}$$

It is the line through $\vec{a}$ parallel to $\vec{b}$, i.e. $\vec{r}=\vec{a}+t\vec{b}$.

Grade - Low

Practice map, main concept: Sheet: J-I 12

Also used in: none

Question 38
How to find a line through a given point (or with a given direction) that intersects two given lines?

Model Answer

Take general points $P(t)$ on L_1 and $Q(s)$ on L_2. Through a point A: A, P, Q collinear ($\overrightarrow{AP} \parallel \overrightarrow{AQ}$). Given direction $\vec{d}$: $\overrightarrow{PQ} \parallel \vec{d}$. Solve for t, s.

Same idea: distance of a point from a line measured parallel to another line = distance to the meeting point.

Grade - Medium

Practice map, main concept: Sheet: J-M 85 | Question Bank: 42, 67

Also used in: none

Dot Product

Question 39
What is the dot product and what are its properties?

Model Answer

$$\vec{a}\cdot\vec{b}=|\vec{a}||\vec{b}|\cos\theta=a_1b_1+a_2b_2+a_3b_3$$

$\hat{i}\cdot\hat{i}=1$, $\hat{i}\cdot\hat{j}=0$; commutative, distributive; $\vec{a}\cdot\vec{a}=|\vec{a}|^2$; $|\vec{a}\pm\vec{b}|^2=|\vec{a}|^2+|\vec{b}|^2\pm2\vec{a}\cdot\vec{b}$.
Ex: $\vec{a}+\vec{b}+\vec{c}=\vec{0}$, $|\vec{a}|=3, |\vec{b}|=5, |\vec{c}|=7$: $\vec{a}\cdot\vec{b}=\frac{49-9-25}{2}=\frac{15}{2}$, angle $\frac{\pi}{3}$.

Grade - High

Practice map, main concept: Sheet: J-I 5; J-II 6-7, 10; J-M 29, 49-50, 98; J-A 18; C-I 15, 23; C-II 33-35 | DPP: 3(1, 3-4, 8, 10); 4(6, 13); 5(3); 8(16) | Question Bank: 5, 7, 49

Also used in: Sheet: J-I 8-9, 11, 29, 31, 34; J-II 17-18, 22, 60; J-M 1-2, 11, 14, 16, 58, 62, 64-65, 67, 91, 103, 106; J-A 1-2, 13-14, 24, 28, 34, 36, 40, 44, 57; C-I 1-2, 7, 16; C-II 15, 23, 29, 36, 40-42, 44-45 | DPP: 2(2, 7); 4(1-3, 7, 10-11); 5(5-6, 11) | Question Bank: 16, 19, 25, 34-35, 37, 43, 48, 59

Question 40
How to find the angle between two vectors using dot product?

Model Answer

$$\cos\theta=\frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$$

$\vec{a}\perp\vec{b} \iff \vec{a}\cdot\vec{b}=0$; acute $\iff \vec{a}\cdot\vec{b}>0$; obtuse $\iff \vec{a}\cdot\vec{b}<0$ (non-parallel).
Ex: $\hat{i}+\hat{j}$ and $\hat{j}+\hat{k}$: $\cos\theta=\frac{1}{2}$, $\theta=60°$.

Grade - High

Practice map, main concept: Sheet: J-I 6-7, 9, 16; J-II 5, 8-9; J-M 11-12, 14, 99, 101; J-A 28; C-I 1, 19; C-II 8, 24, 28, 40, 44 | DPP: 1(9-10); 2(1-2, 4); 3(5, 7); 4(1, 9-11); 5(5-6); 9(2) | Question Bank: 16, 32, 47-48, 65

Also used in: Sheet: J-II 50, 66; J-M 3, 9, 24, 32-33, 55-56, 94; J-A 16, 21-22, 41; C-I 5, 10, 14; C-II 7, 30, 43 | DPP: 1(4); 3(1, 8, 11); 4(12-13); 8(4) | Question Bank: 4, 7, 50, 68

Question 41

How to find the projection of a vector on another, and its components along and perpendicular to it?

Model Answer

$$\text{proj} = \vec{a}\cdot\hat{b}, \quad \vec{a}_{\parallel} = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\vec{b}, \quad \vec{a}_{\perp} = \vec{a} - \vec{a}_{\parallel}$$

Ex: $\vec{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ on $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$: projection $\frac{10}{\sqrt{6}}$, $\vec{a}_{\parallel} = \frac{5}{3}(\hat{i} + 2\hat{j} + \hat{k})$.

Grade - Medium

Practice map, main concept: Sheet: J-I 10; J-M 2, 15, 31, 60, 93; J-A 34, 38, 41; C-I 5 | DPP: 2(5); 4(7) | Question Bank: 1, 4, 22, 24, 51, 58

Also used in: Sheet: J-I 51; J-II 11; J-M 33, 50-51, 54, 59, 92, 95, 97; C-I 13, 23; C-II 29 | DPP: 6(14)

Question 42

How to find the foot of perpendicular and image of a point in a line in space?

Model Answer

Take the general point Q of the line, put $\overrightarrow{PQ}\cdot\vec{b} = 0$ to get t; foot $= Q$, image $= 2Q - P$.

Ex: $P(1, 6, 3)$, line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$: $t = 1$, foot $(1, 3, 5)$, image $(1, 0, 7)$.

Grade - Medium

Practice map, main concept: Sheet: J-II 28, 37; J-M 18-19, 21, 76, 89, 112; J-A 16; C-II 4 | DPP: 3(9); 7(14) | Question Bank: 8, 13, 15, 27, 31, 52, 61, 68

Also used in: Sheet: J-II 35; J-A 3 | Question Bank: 10

Question 43

How to find the perpendicular distance of a point from a line in space?

Model Answer

$$d = \frac{|(\vec{p} - \vec{a}) \times \vec{b}|}{|\vec{b}|}$$

Or $d = PQ$ with Q the foot. Ex: above, $d = \sqrt{13}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 36, 68; J-M 41-42, 78, 110; C-II 9 | DPP: 3(2); 5(10)

Also used in: Sheet: J-II 37; J-M 47, 112; J-A 22; C-II 47 | DPP: 8(8) | Question Bank: 8, 23, 27

Cross Product

Question 44

What is the cross product and what are its properties?

Model Answer

$$\vec{a}\times\vec{b} = |\vec{a}||\vec{b}|\sin\theta\,\hat{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$\hat{i}\times\hat{j} = \hat{k}$ (cyclic), $\vec{b}\times\vec{a} = -\vec{a}\times\vec{b}$, $\vec{a}\times\vec{b} = \vec{0} \iff \vec{a} \parallel \vec{b}$; unit vectors $\perp$ both: $\pm\dfrac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}$.

Ex: $(\hat{i}+\hat{j})\times(\hat{j}+\hat{k})=\hat{i}-\hat{j}+\hat{k}$.

Grade - High

Practice map, main concept: Sheet: J-I 15, 28-29; J-II 15, 17-18, 53, 59-60, 62; J-M 1, 4, 13, 55, 64, 95, 97, 105; J-A 2, 15, 21, 44; C-II 23 | DPP: 4(5, 12); 5(4, 7-8); 7(11); 8(8, 15); 10(12) | Question Bank: 25, 35, 55

Also used in: Sheet: J-I 13-14, 25, 40, 47; J-II 13-14, 29, 34, 55, 58, 61, 66; J-M 8, 15, 22, 60, 80, 93-94, 101, 108, 119; J-A 9, 11, 18, 25, 27-28, 32, 34, 48, 52-53, 57; C-I 2, 5, 8, 10, 17, 19; C-II 8, 17, 19, 45 | DPP: 5(1, 9); 6(8); 8(1, 5, 10-11, 13); 9(4, 10); 10(3-4) | Question Bank: 5, 22, 30, 34, 40, 47, 49, 64, 66

Question 45

What is Lagrange's identity?

Model Answer

$$|\vec{a}\times\vec{b}|^2+(\vec{a}\cdot\vec{b})^2=|\vec{a}|^2|\vec{b}|^2$$

Ex: $|\vec{a}|=2,\ |\vec{b}|=3,\ \vec{a}\cdot\vec{b}=4\Rightarrow|\vec{a}\times\vec{b}|=2\sqrt{5}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 14, 34; J-M 7, 32, 62, 106; J-A 25, 36; C-I 20 | DPP: 5(2) | Question Bank: 20, 34, 39, 50

Also used in: Sheet: J-I 7; J-M 49-50, 56, 91, 98, 101; C-I 12; C-II 8 | DPP: 5(4, 10); 8(5, 15); 9(7) | Question Bank: 48, 55, 59, 65

Question 46

How to find area of triangle, parallelogram and quadrilateral using cross product?

Model Answer

$$\Delta=\tfrac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}|=\tfrac{1}{2}|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}|$$

Parallelogram: $|\vec{a}\times\vec{b}|$ (sides) $=\tfrac{1}{2}|\vec{d}_1\times\vec{d}_2|$ (diagonals); quadrilateral: $\tfrac{1}{2}|\vec{d}_1\times\vec{d}_2|$.

Ex: $A(1,1,1),B(1,2,3),C(2,3,1)$: $\overrightarrow{AB}\times\overrightarrow{AC}=(-4,2,-1)$, area $\frac{\sqrt{21}}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 13; J-II 12-14, 16, 50, 69; J-M 56, 58; J-A 35; C-II 11, 21, 32 | DPP: 5(9); 8(11, 13) | Question Bank: 9, 38, 40, 62

Also used in: Sheet: J-II 31, 33, 40, 57, 68; J-M 78, 120; J-A 12, 38, 41, 54; C-I 21; C-II 31 | DPP: 1(10); 5(7); 9(1) | Question Bank: 15, 17-18, 57-58

Direction Cosines and Direction Ratios

Question 47

What are direction cosines and direction ratios and how are they related?

Model Answer

$$l=\cos\alpha,\ m=\cos\beta,\ n=\cos\gamma,\quad l^2+m^2+n^2=1$$

DRs a,b,c $(\propto l,m,n)\Rightarrow l=\pm\dfrac{a}{\sqrt{a^2+b^2+c^2}}$ etc.; DRs of AB: x_2-x_1,y_2-y_1,z_2-z_1; $\sin^2\alpha+\sin^2\beta+\sin^2\gamma=2$.

Ex: DRs $2,-1,-2\Rightarrow$ DCs $\frac{2}{3},-\frac{1}{3},-\frac{2}{3}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 36; J-M 51, 75 | DPP: 6(14) | Question Bank: 12

Also used in: Sheet: J-I 37, 41; J-II 23; J-M 39, 113, 123; C-I 18 | DPP: 7(5, 12) | Question Bank: 36, 52

Question 48

How to find the angle between two lines from their direction cosines / ratios?

Model Answer

$$\cos\theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Perpendicular: $a_1a_2 + b_1b_2 + c_1c_2 = 0$; parallel: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$. Ex: two diagonals of a cube: $\cos\theta = \frac{1}{3}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 37-38; J-II 23, 26, 57; J-M 39, 72, 123-124 | DPP: 3(11); 7(5) | Question Bank: 6, 14

Also used in: Sheet: J-M 21, 75, 121 | DPP: 4(4)

Straight Line in 3D

Question 49

What is the symmetric (cartesian) form of a line and its general point?

Model Answer

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} = t$$

General point $(x_1 + at,\ y_1 + bt,\ z_1 + ct)$; two-point form uses DRs $x_2 - x_1, \ldots$; a line $\perp$ to two lines has DRs along $\vec{b}_1 \times \vec{b}_2$.

Ex: through $(1, 2, 3)$, DRs $2, -1, 4$: $\frac{x-1}{2} = \frac{y-2}{-1} = \frac{z-3}{4}$, point $(1 + 2t, 2 - t, 3 + 4t)$.

Grade - Medium

Practice map, main concept: Sheet: J-I 35; J-II 25, 29, 58; J-M 120 | DPP: 6(8); 7(2) | Question Bank: 11, 17, 30, 41, 54, 69

Also used in: Sheet: J-I 43; J-II 26-27, 31, 51; J-M 17, 20, 23, 27, 72, 87; J-A 50; C-II 2, 4 | DPP: 3(2); 7(3) | Question Bank: 6, 13, 26, 28, 42, 52

Question 50

How to find the shortest distance between two lines and the condition for them to intersect?

Model Answer

$$SD = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}, \qquad \text{parallel: } \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}$$

Lines intersect (coplanar) $\iff (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$, i.e. $\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0.$

Ex: $(1, 2, 3) + t(1, -3, 2)$, $(4, 5, 6) + s(2, 3, 1)$: $\vec{b}_1 \times \vec{b}_2 = (-9, 3, 9)$, $SD = \frac{3}{\sqrt{19}}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 39; J-II 30, 54, 64; J-M 10, 17, 20, 25, 68, 77, 86-87, 109; J-A 5, 10, 32, 53; C-I 22 | DPP: 6(11); 7(1, 6) | Question Bank: 3, 10, 26

Also used in: Sheet: J-II 39, 67-68; J-M 53, 115; J-A 47; C-I 20; C-II 9, 12 | DPP: 1(8); 10(10)

Plane

Question 51

What are the normal and general forms of the equation of a plane?

Model Answer

$$(\vec{r}-\vec{a})\cdot\vec{n}=0, \quad \vec{r}\cdot\hat{n}=d, \quad ax+by+cz+d=0$$

(a,b,c) = normal; through (x_1,y_1,z_1): $a(x-x_1)+b(y-y_1)+c(z-z_1)=0$; $z=k$ is parallel to the xy-plane.

Ex: through $(1,2,3)$, normal $(2,-1,1)$: $2x-y+z=3$.

Grade - Medium

Practice map, main concept: Sheet: J-M 22, 80, 108; J-A 11, 27, 49 | DPP: 8(14)

Also used in: Sheet: J-I 47; J-II 55, 65; J-M 18, 69; J-A 30, 43, 51; C-II 46 | DPP: 6(2, 10, 17); 7(15)

Question 52

How to find a plane through three points, or through a point parallel to two vectors?

Model Answer

Normal $\vec{n}=\overrightarrow{AB}\times\overrightarrow{AC}$ (or $\vec{b}\times\vec{c}$); parametric form $\vec{r}=\vec{a}+\lambda\vec{b}+\mu\vec{c}$; determinant form $[\vec{r}-\vec{a}\ \ \vec{b}\ \ \vec{c}]=0$.

Ex: through $(1,1,0),(1,2,1),(-2,2,-1)$: $2x+3y-3z=5$.

Grade - Medium

Practice map, main concept: Sheet: J-II 31, 35, 41; J-M 24, 116; J-A 22 | DPP: 6(1, 4-5, 10, 12)

Also used in: Sheet: J-II 36, 40; J-M 63, 66; J-A 46

Question 53

What is the intercept form of a plane?

Model Answer

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$

Meets the axes at $(a,0,0),(0,b,0),(0,0,c)$. Ex: intercepts $2,3,4$: $6x+4y+3z=12$.

Grade - Medium

Practice map, main concept: Sheet: J-I 42; J-II 33; C-I 11; C-II 1, 5 | DPP: 6(2)

Also used in: Sheet: J-II 41; J-M 111, 116; C-II 3 | DPP: 8(12)

Question 54

How to find the angle between two planes?

Model Answer

$$\cos\theta=\frac{|\vec{n}_1\cdot\vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$$

Parallel: $\vec{n}_1\parallel\vec{n}_2$; perpendicular: $\vec{n}_1\cdot\vec{n}_2=0$. Ex: $x+y+2z=9$, $2x-y+z=15$: $\cos\theta=\frac{1}{2}$, $60°$.

Grade - Medium

Practice map, main concept: Sheet: J-M 69 | DPP: 6(9); 9(4); 10(2)

Also used in: Sheet: J-I 46; J-II 36, 44; J-M 34, 73-74, 79, 111; J-A 30

Question 55

How to find the angle between a line and a plane, and when is the line parallel to or in the plane?

Model Answer

$$\sin\theta = \frac{|\vec{b}\cdot\vec{n}|}{|\vec{b}||\vec{n}|}$$

Line $\parallel$ plane: $\vec{b}\cdot\vec{n}=0$; line in plane: also one point of the line satisfies the plane; line $\perp$ plane: $\vec{b}\parallel\vec{n}$.

Ex: DRs $(2,3,6)$, plane $10x+2y-11z=3$: $\sin\theta=\frac{8}{21}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 43, 51; J-II 51; J-M 44, 82, 119; C-I 21 | DPP: 6(6, 13, 15); 7(3)

Also used in: Sheet: J-II 25; J-M 21, 24, 38, 113; J-A 45 | DPP: 7(9); 10(12)

Question 56

How to find the distance of a point from a plane and between parallel planes?

Model Answer

$$d=\frac{|ax_1+by_1+cz_1+d|}{\sqrt{a^2+b^2+c^2}},\qquad d=\frac{|d_1-d_2|}{\sqrt{a^2+b^2+c^2}}$$

Make the coefficients of x,y,z equal before using the second formula. Ex: $x+2y-2z+1=0$, $2x+4y-4z+5=0$: $d=\frac{1}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 47; J-II 55; J-M 118; J-A 43; C-II 3 | DPP: 6(3); 7(15)

Also used in: Sheet: J-I 42, 44, 48; J-II 43-45, 68; J-M 26, 36, 48, 53, 76, 82, 104, 107-108; J-A 4, 19, 22, 30, 47, 49; C-I 11; C-II 1, 5, 9, 47 | DPP: 6(4, 12); 7(8)

Question 57

How to find the foot of perpendicular and image of a point in a plane?

Model Answer

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}=-\frac{ax_1+by_1+cz_1+d}{a^2+b^2+c^2}\quad\text{(image: }-2\times\text{)}$$

Ex: $(1,3,4)$ in $2x-y+z+3=0$: foot $(-1,4,3)$, image $(-3,5,2)$.

Grade - Medium

Practice map, main concept: Sheet: J-I 44, 49; J-II 40; J-M 46, 70, 81, 84, 88, 90, 114; J-A 31, 37, 46 | DPP: 6(17); 7(13); 10(7)

Also used in: Sheet: J-I 50; J-II 46; J-M 73, 117-119; J-A 8, 20, 23, 54; C-I 14 | DPP: 7(16)

Question 58

How to check whether two points are on the same or opposite sides of a plane?

Model Answer

Put both points in $P=ax+by+cz+d$: same sign $\Rightarrow$ same side, opposite signs $\Rightarrow$ opposite sides.

Ex: $(1,1,1)$, $(-1,0,0)$ for $x+y+z-1$: 2 and -2, opposite sides.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-M 71

Question 59

How to find the angle bisector planes of two planes?

Model Answer

$$\frac{a_1x+b_1y+c_1z+d_1}{\sqrt{a_1^2+b_1^2+c_1^2}}=\pm\frac{a_2x+b_2y+c_2z+d_2}{\sqrt{a_2^2+b_2^2+c_2^2}}$$

With $d_1, d_2 > 0$: '+' gives the bisector of the angle containing the origin; it is the acute one if $a_1a_2+b_1b_2+c_1c_2<0$.

Ex: $2x-y+2z+3=0$, $3x-2y+6z+8=0$: origin bisector $5x-y-4z-3=0$.

Grade - Medium

Practice map, main concept: Sheet: J-M 43

Also used in: none

Question 60

How to find where a line meets a plane, and the distance of a point from a plane along a given line?

Model Answer

Put the general point of the line in the plane, solve for t. For distance along $\vec{d}$: line through P parallel to $\vec{d}$, meet the plane at Q, distance PQ.

Ex: $\frac{x-2}{3}=\frac{y+1}{4}=\frac{z-2}{12}$ meets $x-y+z=5$ at $(2,-1,2)$; its distance from $(-1,-5,-10)$ is 13.

Grade - Medium

Practice map, main concept: Sheet: J-I 41, 45, 50; J-II 32; J-M 27, 37, 45, 117; J-A 3, 50; C-II 2 | DPP: 6(16); 7(7, 10)

Also used in: Sheet: J-M 23, 70, 122, 124; J-A 35, 54

Family of Planes, Line of Intersection, Images

Question 61

How to write the family of planes through the line of intersection of two planes?

Model Answer

$$P_1+\lambda P_2=0$$

Find λ from the extra condition (point, parallel / perpendicular, distance). Ex: through $(1,1,1)$ for $x+y+z=1$, $2x+3y+4z=5$: $\lambda=-\frac{1}{2}$, plane $y+2z=3$.

Grade - Medium

Practice map, main concept: Sheet: J-I 46; J-II 38, 44-45; J-M 34-36, 38, 40, 57, 71, 73-74, 79, 111, 121; J-A 4, 19

Also used in: Sheet: J-I 48

Question 62

How to convert the line of intersection of two planes into symmetric form?

Model Answer

Direction $\vec{n}_1\times\vec{n}_2$; a point: put $z=0$ (or a convenient value) and solve the two equations.

Ex: $x+y+z=1$, $2x+3y+4z=5$: direction $(1,-2,1)$, point $(-2,3,0)$: $\frac{x+2}{1}=\frac{y-3}{-2}=\frac{z}{1}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 27, 42, 65, 70; J-M 66, 94, 113; J-A 30, 45, 54; C-I 10; C-II 47 | DPP: 7(4, 9, 12)

Also used in: Sheet: J-I 38; J-II 28, 32, 35, 38, 47; J-M 19, 41-42, 78, 110, 121-122; J-A 20

Question 63

How to find a plane containing a given line and a point, two given lines, or a line and parallel to another line?

Model Answer

Normal $= \vec{b} \times \overrightarrow{AP}$ (line + point P); $\vec{b}_1 \times \vec{b}_2$ (two intersecting lines, or line $\parallel$ another line); $\vec{b} \times \overrightarrow{A_1A_2}$ (two parallel lines). Plane passes through a point A of the line.

Ex: containing $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z}{4}$ and the origin: normal $(4, 4, -5)$, plane $4x + 4y - 5z = 0$.

Grade - Medium

Practice map, main concept: Sheet: J-I 40, 48; J-II 34, 39, 43, 52; J-M 26, 48, 53, 104, 107, 115, 122; J-A 23, 47 | DPP: 7(8)

Also used in: Sheet: J-II 30, 36-37, 57, 67; J-M 84; J-A 5; C-I 22

Question 64

How to find the image (or projection) of a line in a plane, of a line in a line, and of a plane in a plane?

Model Answer

Take two points of the object (for a line meeting the plane: the meeting point + one more point), find their images (or feet) and join them; image of a plane: images of three points, or use $\vec{n}$ reflected.

Line $\parallel$ plane: the image is parallel to it, through the image of one point.

Grade - Medium

Practice map, main concept: Sheet: J-II 46; J-M 47; J-A 8, 20 | DPP: 7(16)

Also used in: none

Question 65

What is the line of greatest slope and how is its direction found?

Model Answer

$$\vec{d} = (\vec{n}_H \times \vec{n}_G) \times \vec{n}_G$$

A line in the given plane G perpendicular to its line of intersection with the horizontal plane H. Ex: $H : 4x - 3y + 7z = 0$, $G : 2x + y - 5z = 0$, through $(2, 1, 1)$: DRs $3, -1, 1$.

Grade - Low

Practice map, main concept: Sheet: J-II 47

Also used in: none

Tetrahedron and Parallelepiped

Question 66

What are the important properties of a tetrahedron and a parallelepiped?

Model Answer

Tetrahedron: 4 vertices, 6 edges (3 pairs of opposite edges), 4 faces; centroid $\dfrac{\vec{a} + \vec{b} + \vec{c} + \vec{d}}{4}$ = meet of lines joining mid-points of opposite edges; height = projection of a lateral edge on the normal of the base.

Regular tetrahedron: all faces equilateral, opposite edges perpendicular, angle between faces $\cos^{-1} \frac{1}{3}$.
Parallelepiped: 6 parallelogram faces, centre = mid-point of any diagonal.

Grade - Medium

Practice map, main concept: Sheet: C-II 6, 12 | Question Bank: 18

Also used in: Sheet: J-II 20-21; C-I 18; C-II 3, 32 | DPP: 8(12); 10(10) | Question Bank: 9

Scalar Triple Product

Question 67

What is the scalar triple product and what are its properties?

Model Answer

$$[\vec{a}\ \vec{b}\ \vec{c}] = \vec{a}\cdot(\vec{b}\times\vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Cyclic: $[\vec{a}\vec{b}\vec{c}] = [\vec{b}\vec{c}\vec{a}]$, swap $\Rightarrow$ sign changes; dot and cross can be interchanged; two equal (parallel) vectors $\Rightarrow 0$; $[\vec{a}+\vec{b}\ \ \vec{b}+\vec{c}\ \ \vec{c}+\vec{a}] = 2[\vec{a}\vec{b}\vec{c}]$; $[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}] = [\vec{a}\vec{b}\vec{c}]^2$.

Ex: $[(1,2,3),(0,1,2),(1,0,1)] = 2$.

Grade - Medium

Practice map, main concept: Sheet: J-I 18, 23, 27; J-M 16, 67; J-A 12, 51; C-I 3, 8; C-II 10 | DPP: 8(1, 4-6, 10); 9(5, 10); 10(6) | Question Bank: 33, 43

Also used in: Sheet: J-I 20, 54; J-II 59, 64; J-M 3, 5-6, 52, 105; J-A 6-7, 56; C-II 14, 16, 27, 37-39 | DPP: 6(12); 8(3, 9, 15); 10(1, 3, 9, 11) | Question Bank: 66

Question 68

How to evaluate the product of two scalar triple products, and $[\vec{a}\ \vec{b}\ \vec{c}]^2$?

Model Answer

$$[\vec{a}\vec{b}\vec{c}][\vec{p}\vec{q}\vec{r}] = \begin{vmatrix} \vec{a}\cdot\vec{p} & \vec{a}\cdot\vec{q} & \vec{a}\cdot\vec{r} \\ \vec{b}\cdot\vec{p} & \vec{b}\cdot\vec{q} & \vec{b}\cdot\vec{r} \\ \vec{c}\cdot\vec{p} & \vec{c}\cdot\vec{q} & \vec{c}\cdot\vec{r} \end{vmatrix}$$

Put $\vec{p},\vec{q},\vec{r} = \vec{a},\vec{b},\vec{c}$ for $[\vec{a}\vec{b}\vec{c}]^2$. Ex: unit vectors with mutual angles $60°$: $[\vec{a}\vec{b}\vec{c}]^2 = \frac{1}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 19; J-A 40; C-I 6

Also used in: Sheet: J-I 21

Question 69

How to find the volume of a parallelepiped and of a tetrahedron?

Model Answer

$$V_{\text{par}} = |[\vec{a}\ \vec{b}\ \vec{c}]|, \qquad V_{\text{tet}} = \tfrac{1}{6}|[\overrightarrow{AB}\ \overrightarrow{AC}\ \overrightarrow{AD}]|$$

Also $V_{\text{tet}} = \frac{1}{3}(\text{base area})(\text{height})$. Ex: vertices $O,(1,0,0),(0,2,0),(0,0,3)$: $V = 1$.

Grade - Medium

Practice map, main concept: Sheet: J-I 21; J-II 20-21; J-A 6; C-I 17; C-II 26, 31, 39, 43 | DPP: 8(3, 12); 9(1); 10(3, 10)

Also used in: Sheet: J-II 36, 41; J-M 116; C-I 3, 20; C-II 6, 9, 12 | DPP: 9(5) | Question Bank: 18

Question 70

How to test coplanarity of three vectors, four points or two lines?

Model Answer

Three vectors coplanar (dependent) $\iff [\vec{a}\vec{b}\vec{c}] = 0$; points $A, B, C, D \iff [\overrightarrow{AB}\ \overrightarrow{AC}\ \overrightarrow{AD}] = 0$; lines $\iff [\vec{a}_2 - \vec{a}_1\ \ \vec{b}_1\ \ \vec{b}_2] = 0$.

Ex: $\hat{i} + \hat{j} + \hat{k}$, $\hat{i} + 2\hat{j} + 3\hat{k}$, $2\hat{i} + 3\hat{j} + 4\hat{k}$: determinant 0, coplanar.

Grade - Medium

Practice map, main concept: Sheet: J-I 20, 24, 54; J-II 66; J-M 5-6, 65, 102; J-A 7, 48, 56; C-I 18 | DPP: 8(2, 7, 9); 9(8); 10(4)

Also used in: Sheet: J-I 23; J-II 70; C-II 20 | DPP: 7(4); 10(5-6)

Vector Triple Product

Question 71

What is the vector triple product formula and how is it used?

Model Answer

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$$

It lies in the plane of $\vec{b}, \vec{c}$ and is $\perp \vec{a}$; $(\vec{a} \times \vec{b}) \times \vec{c} \neq \vec{a} \times (\vec{b} \times \vec{c})$ in general. A vector coplanar with $\vec{p}, \vec{q}$ and $\perp \vec{r}$: $\lambda\, \vec{r} \times (\vec{p} \times \vec{q})$.

Ex: $\hat{i} \times (\hat{i} \times \hat{j}) = -\hat{j}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 11, 26, 30-33; J-II 22; J-M 8-9, 30, 52, 96, 100; J-A 13; C-I 4, 24; C-II 13, 38, 46 | DPP: 9(7, 9); 10(8, 11, 13) | Question Bank: 59

Also used in: Sheet: J-I 17, 52-53; J-II 19, 60; J-M 4, 28; C-I 9, 23; C-II 14, 39 | DPP: 5(2); 9(6) | Question Bank: 2, 63

Question 72

How to simplify products of four vectors?

Model Answer

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{a} \cdot \vec{d})(\vec{b} \cdot \vec{c})$$

$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a}\vec{b}\vec{d}]\vec{c} - [\vec{a}\vec{b}\vec{c}]\vec{d} = [\vec{a}\vec{c}\vec{d}]\vec{b} - [\vec{b}\vec{c}\vec{d}]\vec{a}$$

Grade - Medium

Practice map, main concept: Sheet: J-I 25; J-II 61; J-M 103; J-A 29; C-I 12; C-II 14

Also used in: Sheet: J-II 69

Linear Combination in Space and Vector Equations

Question 73

How to express a vector as a linear combination of three non-coplanar vectors?

Model Answer

$$\vec{r} = x\vec{a} + y\vec{b} + z\vec{c}, \quad x = \frac{[\vec{r}\,\vec{b}\,\vec{c}]}{[\vec{a}\,\vec{b}\,\vec{c}]},\ y = \frac{[\vec{a}\,\vec{r}\,\vec{c}]}{[\vec{a}\,\vec{b}\,\vec{c}]},\ z = \frac{[\vec{a}\,\vec{b}\,\vec{r}]}{[\vec{a}\,\vec{b}\,\vec{c}]}$$

(Dot with $\vec{b} \times \vec{c}$ etc.; or compare components.) Reciprocal system: $\vec{a}' = \dfrac{\vec{b} \times \vec{c}}{[\vec{a}\vec{b}\vec{c}]}$, $\vec{a} \cdot \vec{a}' = 1$.

Ex: $2\hat{i} + 3\hat{j} + 4\hat{k}$ with $\hat{i}$, $\hat{i} + \hat{j}$, $\hat{i} + \hat{j} + \hat{k}$: $x = -1,\ y = -1,\ z = 4$.

Grade - Medium

Practice map, main concept: Sheet: J-A 14, 17, 52; C-II 16-17 | DPP: 10(1, 9)

Also used in: Sheet: J-M 30 | Question Bank: 21

Question 74

How to solve a vector equation?

Model Answer

Do not introduce new vectors; isolate the unknown; take dot or cross product of both sides with a known vector, then use VTP.

Ex: $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$, $\vec{r} \cdot \vec{c} = 0 \Rightarrow \vec{r} = \vec{a} + t\vec{b}$, $t = -\dfrac{\vec{a} \cdot \vec{c}}{\vec{b} \cdot \vec{c}}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 17, 52-53; J-II 19; J-M 28, 59, 92; J-A 57; C-I 2, 9; C-II 19, 37 | DPP: 5(1); 9(6) | Question Bank: 2, 63-64, 66

Also used in: Sheet: J-I 28; J-A 44; C-I 4 | DPP: 10(13)

Miscellaneous

Question 75

How to use vectors for geometric proofs and maximum / minimum problems?

Model Answer

Expand squares of magnitudes with dot products and use $|\vec{v}|^2 \geq 0$, $|\cos\theta| \leq 1$, $|\sin\theta| \leq 1$ or AM-GM.

Ex: unit $\vec{a}, \vec{b}, \vec{c}$: $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9 - |\vec{a} + \vec{b} + \vec{c}|^2 \leq 9$.

Grade - Medium

Practice map, main concept: Sheet: J-I 8; J-A 1, 24, 55; C-I 7; C-II 29, 36, 42 | DPP: 5(11)

Also used in: Sheet: J-II 7, 12; C-II 24, 26 | DPP: 1(2); 3(7); 8(16) | Question Bank: 39